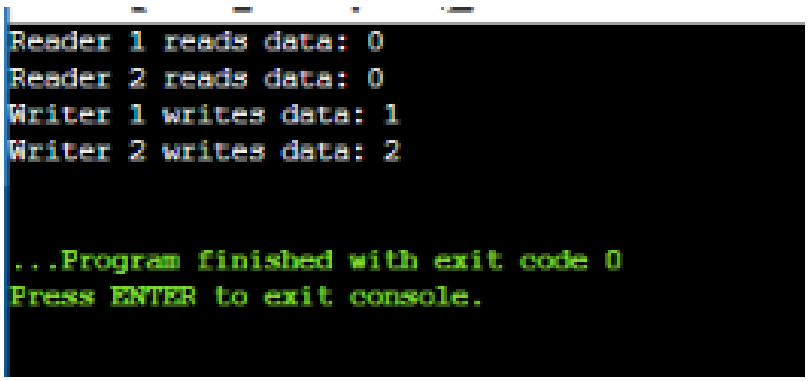
EXPERIMENT 20: Reader-Writer Problem Using Semaphores

code:

```
#include <stdio.h>
#include <pthread.h>
#include <semaphore.h>
#include <unistd.h>

sem_t mutex, writeBlock;

int data = 0;
int readersCount = 0;

void *reader(void *arg) {

   int id = *(int *)arg;

   sem_wait(&mutex);

   readersCount++;

   if (readersCount == 1)
      sem_wait(&writeBlock);

   sem_post(&mutex);

   printf("Reader %d reads data: %d\n", id, data);

   sleep(1);

   sem_wait(&mutex);
```

```
    readersCount--;

    if (readersCount == 0)
        sem_post(&writeBlock);

    sem_post(&mutex);

    return NULL;
}

void *writer(void *arg) {

    int id = *(int *)arg;

    sem_wait(&writeBlock);

    data++;

    printf("Writer %d writes data: %d\n", id, data);

    sleep(1);

    sem_post(&writeBlock);

    return NULL;
}

int main() {

    pthread_t r1, r2, w1, w2;
```

```
    int a = 1, b = 2;

    sem_init(&mutex, 0, 1);
    sem_init(&writeBlock, 0, 1);

    pthread_create(&r1, NULL, reader, &a);
    pthread_create(&r2, NULL, reader, &b);

    pthread_create(&w1, NULL, writer, &a);
    pthread_create(&w2, NULL, writer, &b);

    pthread_join(r1, NULL);
    pthread_join(r2, NULL);

    pthread_join(w1, NULL);
    pthread_join(w2, NULL);

    sem_destroy(&mutex);
    sem_destroy(&writeBlock);

    return 0;
}
```

Output: